

Chapter 4: Data Input and Output

1. Introduction

Data input is the process of capturing geographic data (both spatial and attribute coordinates) and converting it into a digital format that can be recognized, stored, and processed by a GIS database. This phase is often the most time consuming, labor-intensive, and financially demanding part of a GIS project, frequently accounting for up to **70% of total project costs**. Output, conversely, is the phase where processed spatial information is delivered to users in human readable formats to facilitate decision making.

2. Spatial Data Input Techniques

Depending on the raw data source, several distinct technical methods are used to ingest spatial geometries into a GIS:

- **Manual Digitizing:**
 - **Tablet Digitizer:** A legacy hardware method where a hardcopy paper map is taped to a magnetic tabletop grid, and an operator manually traces points, lines, and polygons using a handheld cursor (puck).
 - **Heads-Up (On-Screen) Digitizing:** The modern standard. An operator imports an Orthorectified raster image (satellite imagery or scanned map) as a background layer on the computer monitor and uses a mouse to trace vector shapes directly over it.
- **Automatic Scanning:** High-resolution optical scanners automatically read paper maps or aerial photographs, converting the physical drawing into a digital raster matrix grid of pixels. This must be followed by image to vector conversion software if vector layers are required.
- **Direct Digital Coordinate Entry (COGO Coordinate Geometry):** Precise land surveying measurements (such as angles, bearings, distances, and azimuths) collected via

Total Stations or digital levels are typed directly into the GIS interface to construct highly accurate engineering or cadastral boundaries.

- **GNSS/GPS Data Ingestion:** Importing digital files directly from field data loggers. The points are plotted using absolute Earth coordinates, removing manual tracing errors.
- **Remote Sensing Data Integration:** Loading pre-processed digital raster datasets directly from space borne or airborne sensors (such as Landsat, Sentinel, or LiDAR point clouds).

3. Spatial Data Editing

Raw spatial data inputs invariably contain geometric errors that must be systematically corrected before any analytical workflows can begin.

Common Digitizing Errors

- **Overshoots:** A line extends beyond its intended intersection point with another line.
- **Undershoots:** A line stops short and fails to connect with an adjacent target line.
- **Sliver Polygons:** Tiny, erroneous gap errors or overlapping gaps created along shared boundaries when adjacent polygons are traced independently instead of using snapping tools.
- **Dangling Nodes:** Unconnected endpoints of lines that should otherwise be joined to a structural network.

The Snapping Process

To eliminate these errors, GIS software relies on **snapping environments**. Snapping automatically pulls a cursor or coordinate point directly to an existing vertex, edge, or end node within a pre-defined search tolerance radius, ensuring absolute mathematical closure.

4. Topology Creation

Topology is the mathematical procedure that explicitly defines the geometric spatial relationships between features (such as adjacency, connectivity, and containment) regardless of map projections or scale.

Topology does not care about coordinates; it cares about how elements are linked. During topology creation, the software builds a structural database framework that rules:

- **Connectivity (Arc-Node Topology):** Defines how lines intersect at common nodes, which is absolutely vital for downstream highway networks or pipeline routing analysis.
- **Area Definition (Polygon-Arc Topology):** Defines which continuous line segments close together to bound a specific, distinct area polygon.
- **Contiguity (Left-Right Topology):** Explicitly records which polygon lies to the left and which lies to the right of every single arc line segment, allowing the system to understand shared boundaries seamlessly.

5. Structuring Map Data

Structuring map data means organizing the corrected, topological spatial information into a logical system that the GIS software can store and manipulate efficiently.

- **Layering / Thematic Separation:** The real world is separated into independent, structured geographic themes (e.g., separating a city into a soils layer, a roads layer, a hydrology layer, and a boundary parcel layer) rather than storing everything on a single drawing canvas.
- **Linking Spatial and Attribute Data:** Every geometric object is assigned a unique system identifier a **Feature ID (FID)** or **Object ID (OID)**. This ID acts as a primary database key that directly connects the spatial shape on the map to its corresponding descriptive row in a relational database management system table.

6. GIS Data Output

Output is the final subsystem of a GIS, responsible for displaying the results of spatial data manipulation and analysis in formats tailored to specific audiences.

- **Map Production (Visual Cartography):** The generation of professional digital map layouts that adhere to strict cartographic principles, containing essential map elements such as:
 - Scale bars (graphic and verbal)
 - North arrows or coordinate orientation grids
 - Comprehensive legends explaining symbols and color ramps
 - Titles, subtitles, and data source citations
- **Digital Data Export:** Saving files into standard interchangeable geospatial industry formats (such as Esri Shapefiles, GeoJSON, KML/KMZ files for Google Earth, or GeoTIFFs for raster arrays) to allow other systems to access the data.
- **Tabular Summaries and Reports:** Outputting statistical tables, spreadsheets, or attribute lists (e.g., a text document listing all land parcels affected by a proposed road extension project) alongside the visual maps.